Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 1 of 10 · Question 8 part b · 4 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

8 (a)	• provides nutrients/named nutrient/oxygen to body cells; • removes waste/named waste from body cells;		2
(b)	• capillaries have small gaps/pores; • water/small molecules/named small molecule forced out; • due to high (hydrostatic) pressure (in capillaries); • at arteriole end;	ignore plasma allow due to high pressure (in capillaries) allow artery	4
(c)	• osmosis; • lower water potential inside cells/higher water potential in tissue fluid;; • water moves from high to low water potential/down a water potential gradient;		3
(d)	• swelling/increase in cell volume; • cells burst/lyse;		2

Total Marks 11

Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 1 of 10 · Question 8 part b · 4 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

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Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 2 of 10 · Question 4 part a.ii · 4 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number				
4	(a)	(i)	Two from: • no red blood cells; • no plasma/large proteins; • higher oxygen concentration in the blood; • lower carbon dioxide concentration in the blood;	ORA	2
		(ii)	• artery end has a higher pressure; • capillaries have holes/pores/are permeable to small molecules/are leaky; • water forced/passes out; • small molecules are forced out/filtered/larger molecules/named larger molecule remain in the blood;	Allow blood enters capillaries under high pressure	4
		(iii)	• excess tissue fluid drains through the lymph; • muscle contraction moves fluid in the lymphatic system;		2
	(b)		Three from: • antibodies have a variable region/contain receptors; • variable region/receptors have a specific shape/all antibodies are different; • variable region/receptors are a complementary shape to antigen; • antibodies bind to an antigen;		3

Total 11 marks

Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 2 of 10 · Question 4 part a.ii · 4 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 3 of 10 · Question 7 part a.i · 2 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

7 (a) (i)	any two from: • pressure falls because of fall in volume; • fluid drains from capillaries; • and into lymph vessels • further away from heart ; • larger cross sectional area; • increased resistance/friction; • lumen of vein large(er)/artery narrow(er);		2
(ii)	• stay in capillary; • too large to pass out;		2
(b)	• plasma in test tube; • add biuret reagent; • (blue to) lilac/purple; • shows presence of protein;	ORA for last two points linked to mp3	4
(c)	• arterial end 4 - 3.3 = 0.7kPa; • (net) flow out; • venous end 1.6 - 3.3 = -1.7kPa; • (net) flow in;		4
(d)	• volume of blood decreases/less blood; • blood pressure drops;		2
			14 marks

Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 3 of 10 · Question 7 part a.i · 2 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

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Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 4 of 10 · Question 7 part a.ii · 2 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

7 (a) (i)	any two from: • pressure falls because of fall in volume; • fluid drains from capillaries; • and into lymph vessels • further away from heart ; • larger cross sectional area; • increased resistance/friction; • lumen of vein large(er)/artery narrow(er);		2
(ii)	• stay in capillary; • too large to pass out;		2
(b)	• plasma in test tube; • add biuret reagent; • (blue to) lilac/purple; • shows presence of protein;	ORA for last two points linked to mp3	4
(c)	• arterial end 4 - 3.3 = 0.7kPa; • (net) flow out; • venous end 1.6 - 3.3 = -1.7kPa; • (net) flow in;		4
(d)	• volume of blood decreases/less blood; • blood pressure drops;		2
			14 marks

Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 4 of 10 · Question 7 part a.ii · 2 marks

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Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 5 of 10 · Question 7 part c · 4 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

7 (a) (i)	any two from: • pressure falls because of fall in volume; • fluid drains from capillaries; • and into lymph vessels • further away from heart ; • larger cross sectional area; • increased resistance/friction; • lumen of vein large(er)/artery narrow(er);		2
(ii)	• stay in capillary; • too large to pass out;		2
(b)	• plasma in test tube; • add biuret reagent; • (blue to) lilac/purple; • shows presence of protein;	ORA for last two points linked to mp3	4
(c)	• arterial end 4 - 3.3 = 0.7kPa; • (net) flow out; • venous end 1.6 - 3.3 = -1.7kPa; • (net) flow in;		4
(d)	• volume of blood decreases/less blood; • blood pressure drops;		2
			14 marks

Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 5 of 10 · Question 7 part c · 4 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

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Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 6 of 10 · Question 7 part d · 2 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part d in this mark scheme extract

7 (a) (i)	any two from: • pressure falls because of fall in volume; • fluid drains from capillaries; • and into lymph vessels • further away from heart ; • larger cross sectional area; • increased resistance/friction; • lumen of vein large(er)/artery narrow(er);		2
(ii)	• stay in capillary; • too large to pass out;		2
(b)	• plasma in test tube; • add biuret reagent; • (blue to) lilac/purple; • shows presence of protein;	ORA for last two points linked to mp3	4
(c)	• arterial end 4 - 3.3 = 0.7kPa; • (net) flow out; • venous end 1.6 - 3.3 = -1.7kPa; • (net) flow in;		4
(d)	• volume of blood decreases/less blood; • blood pressure drops;		2
			14 marks

Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 6 of 10 · Question 7 part d · 2 marks

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Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 7 of 10 · Question 4 part a.iii · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

Number				
4	(a) (i)	Two from: • no red blood cells; • no plasma/large proteins; • higher oxygen concentration in the blood; • lower carbon dioxide concentration in the blood;	ORA	2
	(ii)	• artery end has a higher pressure; • capillaries have holes/pores/are permeable to small molecules/are leaky; • water forced/passes out; • small molecules are forced out/filtered/larger molecules/named larger molecule remain in the blood;	Allow blood enters capillaries under high pressure	4
	(iii)	• excess tissue fluid drains through the lymph; • muscle contraction moves fluid in the lymphatic system;		2
	(b)	Three from: • antibodies have a variable region/contain receptors; • variable region/receptors have a specific shape/all antibodies are different; • variable region/receptors are a complementary shape to antigen; • antibodies bind to an antigen;		3

Total 11 marks

Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 7 of 10 · Question 4 part a.iii · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 8 of 10 · Question 4 part a.i · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

Number				
4	(a) (i)	Two from: • no red blood cells; • no plasma/large proteins; • higher oxygen concentration in the blood; • lower carbon dioxide concentration in the blood;	ORA	2
	(ii)	• artery end has a higher pressure; • capillaries have holes/pores/are permeable to small molecules/are leaky; • water forced/passes out; • small molecules are forced out/filtered/larger molecules/named larger molecule remain in the blood;	Allow blood enters capillaries under high pressure	4
	(iii)	• excess tissue fluid drains through the lymph; • muscle contraction moves fluid in the lymphatic system;		2
	(b)	Three from: • antibodies have a variable region/contain receptors; • variable region/receptors have a specific shape/all antibodies are different; • variable region/receptors are a complementary shape to antigen; • antibodies bind to an antigen;		3

Total 11 marks

Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 8 of 10 · Question 4 part a.i · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 9 of 10 · Question 7 part d · 3 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part d in this mark scheme extract

Number				
7	(a)	pulmonary artery;		1
	(b)	• Y is thicker; • blood pumped into aorta; • takes/distributes blood to organs/tissues/whole body; • requires high pressure • thicker wall means more muscle for stronger contraction;	ORA for each point	5
	(c)	any three of • heart muscle deprived of oxygen; • insufficient energy for muscle contraction/less aerobic respiration/(more) anaerobic respiration; • heart stops pumping blood/pumps less blood; • heart attack;		3
	(d)	• transfers CO_2 (from muscle cells) to blood capillary; • transfers oxygen from blood/capillary (to muscle cells); • transfers glucose from blood/capillary (to muscle cells);		3

Total 12 marks

Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 9 of 10 · Question 7 part d · 3 marks

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Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 10 of 10 · Question 8 part a · 2 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

8 (a)	• provides nutrients/named nutrient/oxygen to body cells; • removes waste/named waste from body cells;		2
(b)	• capillaries have small gaps/pores; • water/small molecules/named small molecule forced out; • due to high (hydrostatic) pressure (in capillaries); • at arteriole end;	ignore plasma allow due to high pressure (in capillaries) allow artery	4
(c)	• osmosis; • lower water potential inside cells/higher water potential in tissue fluid;; • water moves from high to low water potential/down a water potential gradient;		3
(d)	• swelling/increase in cell volume; • cells burst/lyse;		2

Total Marks 11

Type MS · Explain how tissue fluid forms, what it does and why it builds up

Answer 10 of 10 · Question 8 part a · 2 marks

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